

Linux Programming: Assignment-4:

20-09-2025

NAME- G JYOTHSNA

ROLL NO- 04

USN- ENG24CY0021

CLASS- CYBER SECURITY

SECTION- C

1. A system has a file /etc/passwd. How would you use grep + tee to extract usernames and save them to a file while also displaying them on screen?

Ans

Command:

1. cut -d: -f1 /etc/passwd | tee usernames.txt
2. tee

2. A binary isn't found in \$PATH. How would you use commands (which, find, locate) to troubleshoot and fix the issue?

Ans

Command

1. which binary_name

Returns nothing if binary is no there

2. sudo find / -type f -name binary_name 2>/dev/null

Checks all the directories

3. locate binary_name

Uses indexed database to search which makes it faster

4.export PATH=\$PATH:/path/to/binary

Used to add the path to binary after it is found

3. Write a command pipeline that finds all .log files modified in the last 24 hours in /var/log and saves results into log_report.txt.

Ans

Command

```
find /var/log -type f -name "*.log" -mmin -1440 | tee log_report.txt
```

Here the find commands finds all the log files while the tee commands saves it

4. What is the difference between shutdown -r now and reboot?

Ans

Shutdown -r now: this command first shutdown the system and then restarts it and is usually safe as it doesn't harm any files

Reboot: this command directly reboots the system which sometimes can delete some unsaved data

5. How can you use the tee command to debug a script that generates both standard output and error messages?

Ans

Command

```
./script.sh 2>&1 | tee debug.log
```

It redirects both standard output and errors into debug_output.txt while displaying them.

6. Explain any three real-world applications of Linux in industries

Ans

1.Cloud/DevOps

Google cloud rely heavily on linux Virtual machine for deployment

2. Supercomputing and Scientific Research

Linux dominates the supercomputing world. The majority of the world's top supercomputers run Linux due to its flexibility and performance

3. Embedded Systems in Automotive Industry

Linux is widely used in embedded systems for modern vehicles and for navigation, and advanced driver assistance systems as embedded system mostly require more hardware related language linux is usually preferred

7. Differentiate application, system and utility software in the context of Linux environment.

Ans

System Software: This includes essential software that manages the hardware and creates application software that can run. It acts as a bridge between user commands and the physical components of the computer.

Example: The Linux kernel

Application Software: These are programs created to help users carry out specific tasks on a computer. They are designed with a focus on usability and function for end-users.

Utility Software: Special tools that help in system monitoring, maintenance, and management. Example: grep, tar, top

8. What are the key differences between open-source and proprietary operating systems?

Open-source operating systems: such as Linux, provides free access to their source code. Which allows users to study, modify

the software according to their needs. They are generally free of cost, highly customizable, and maintained by large communities that offer updates and support. Open-source systems are also more transparent, as users can see how the system works internally.

Proprietary operating systems like Windows and macOS do not share their source code with the public. Only the company that owns the software can change or control it. Users usually have to buy a license to use these systems. They are often easy to use and come with official support, but they don't allow much customization. Also, because the code is hidden, users have less control and cannot see how the system works inside.

9. Write the command to display the system's kernel version.

Ans

Command:

`uname -r`

10. What is the difference between head and tail commands in text processing?

Ans

Command

head: this command displays the first 10 lines of the file you want
`head filename.txt`

tail : this command displays the last 10 lines of the file you want
`tail filename.txt`

